# Mikhail Razakov

## Software Engineer — Systems & Architecture

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I design and build software systems, from compilers and program-analysis tools to backend services and offline-first applications. I work on symbolic computation in SharpChecker at ISP RAS and previously implemented LLVM optimization at MCST. In independent projects, I own architecture, implementation, and correctness checks.

**Professional experience**
ISP RAS: SharpChecker and symbolic computation. MCST: LLVM 22/C++ and optimization validation.

**Independent engineering**
UniversalToolchain compiler/runtime; botay! web/Android offline sync; VpnMediator payment-state recovery.

**External results**
Talk delivered at LangDev on 9 Oct 2026; top MCST technical-selection result (organizer correspondence).

## EXPERIENCE

**October 2026 — present**

### ISP RAS — SharpChecker development

- Work on symbolic computation in SharpChecker, a C#/.NET static-analysis tool.

**July — August 2026**

### MCST — Compiler Engineering Intern

- Implemented a LICM pass for LLVM 22 in C++; hoisting conditions check loop structure, memory access, side effects, and speculative safety.
- Built a Python harness comparing program behavior before/after optimization and checking transform/no-transform cases.

## PROJECTS

### UniversalToolchain / Wist2

Resolved component dependencies, conflicts, and backend routes into an immutable LanguagePlan; runtime materializes the checked selection without replanning.

Frontend, IR, optimization passes, interpreter and CIL backend; independent configuration checks and 1,324 tests at the 9 Oct 2026 revision.

### botay! / student-execution-os

Built a web/Android planner separating external observations and user state from derived plans; feasible schedules require a validated witness.

Offline operations replay idempotently; the signed Android 0.8.3 release workflow passed on 10 Oct 2026.

### VpnMediator

Built backend logic for payments and subscriptions, preserving provider evidence independently of access changes.

Idempotent retries, reconciliation and manual review for conflicts, with backup/restore and rollback safeguards; no scale claim.

## SKILLS

**Core languages**
C++23, C#/.NET, Python; also Rust, Java, JavaScript, SQL

**Software / systems**
modular architecture, compiler tooling, APIs, Linux, SQLite/PostgreSQL

**Product / backend**
offline synchronization, Android, payments, state recovery, migrations

**Correctness**
regression and differential tests, LLVM Verifier, invariants, idempotency

## EDUCATION

HSE University — Software Engineering

## RECOGNITION

**MCST LLVM/C++ selection**
Organizer correspondence: only 5.0/5.0 score and only solution passing all 104/104 tests among 49 submissions.

**LangDev'26 — Build the Language, Then Make the Abstractions Disappear**
Delivered the UniversalToolchain architecture talk at LangDev on 9 October 2026.

Moscow / remote